

Dimension Reduction Analysis: pbmc

Generated report #1

Seed: 0 | Plan revision: 1

Dataset profile & preprocessing

Peripheral blood mononuclear cells (PBMCs) from a healthy donor, profiled with 10x Genomics single-cell RNA sequencing (scRNA-seq). This is the standard "pbmc3k" dataset used throughout the Scanpy tutorials and widely used as a benchmark for single-cell analysis methods. Stored as the raw h5ad file from the Scanpy tutorials.

Samples: 2,700 Features: 32,738 Sparsity: 97.4% Missing values: No

Preprocessing: `normalize_total`

Raw UMI counts (`profile.json`: mean ~2367 total counts per cell) have very different per-cell totals; `DATA_DESCRIPTION.md` notes no normalization has been applied yet. Library-size normalization removes sequencing-depth differences between cells before comparing expression profiles.

Preprocessing: `log1p`

Raw counts are heavily right-skewed; `DATA_DESCRIPTION.md` notes a small number of highly-expressed genes can dominate raw totals. Log transform compresses this skew so distance-based methods (PCA, UMAP) aren't dominated by a handful of extreme values.

Preprocessing: `select_hvg`

32,738 genes is far more than 2,700 cells, and 97.4% of entries are zero (`profile.json`), so most genes carry little signal. Selecting the top 2,000 highly variable genes keeps the informative signal and keeps distance computations from being dominated by noise genes.

Methods and hyperparameters

pca (role: general_purpose, status: ok)

Hyperparameters: `n_components=50`

Required linear baseline. 50 components is the standard scRNA-seq choice: enough to capture most biological variance across 2,000 genes without overfitting to per-gene noise given only 2,700 cells.

umap (role: visualization_only, status: ok)

Hyperparameters: `n_neighbors=15`, `min_dist=0.1`, `n_components=2`, `random_state=0`

Standard nonlinear manifold method for scRNA-seq visualization. `n_neighbors=15` and `min_dist=0.1` are UMAP's own defaults; with 2,700 cells there's no strong reason yet to move off the defaults, this is a first-pass plan for pipeline testing, not a final analysis.

Quantitative metrics

method	trustworthiness	variance explained (PCs used)	PCs for 90% variance
pca	0.9344	0.2904	>200
umap	0.7740		

How to read the metrics

Trustworthiness (`sklearn.manifold.trustworthiness`, 10 neighbours)

For each sample, takes its 10 nearest neighbours in the embedding and penalizes any that were not close neighbours in the original preprocessed data, weighting each penalty by how far down the original neighbour ranking that point sat. Ranges 0 to 1; 1 means the embedding introduces no false neighbours. It measures local neighbourhood fidelity only, not preservation of global distances. Computed against the data after the plan's preprocessing steps; on a random 5,000-sample subset when the dataset is larger than that (see Limitations).

Variance explained (PCs used)

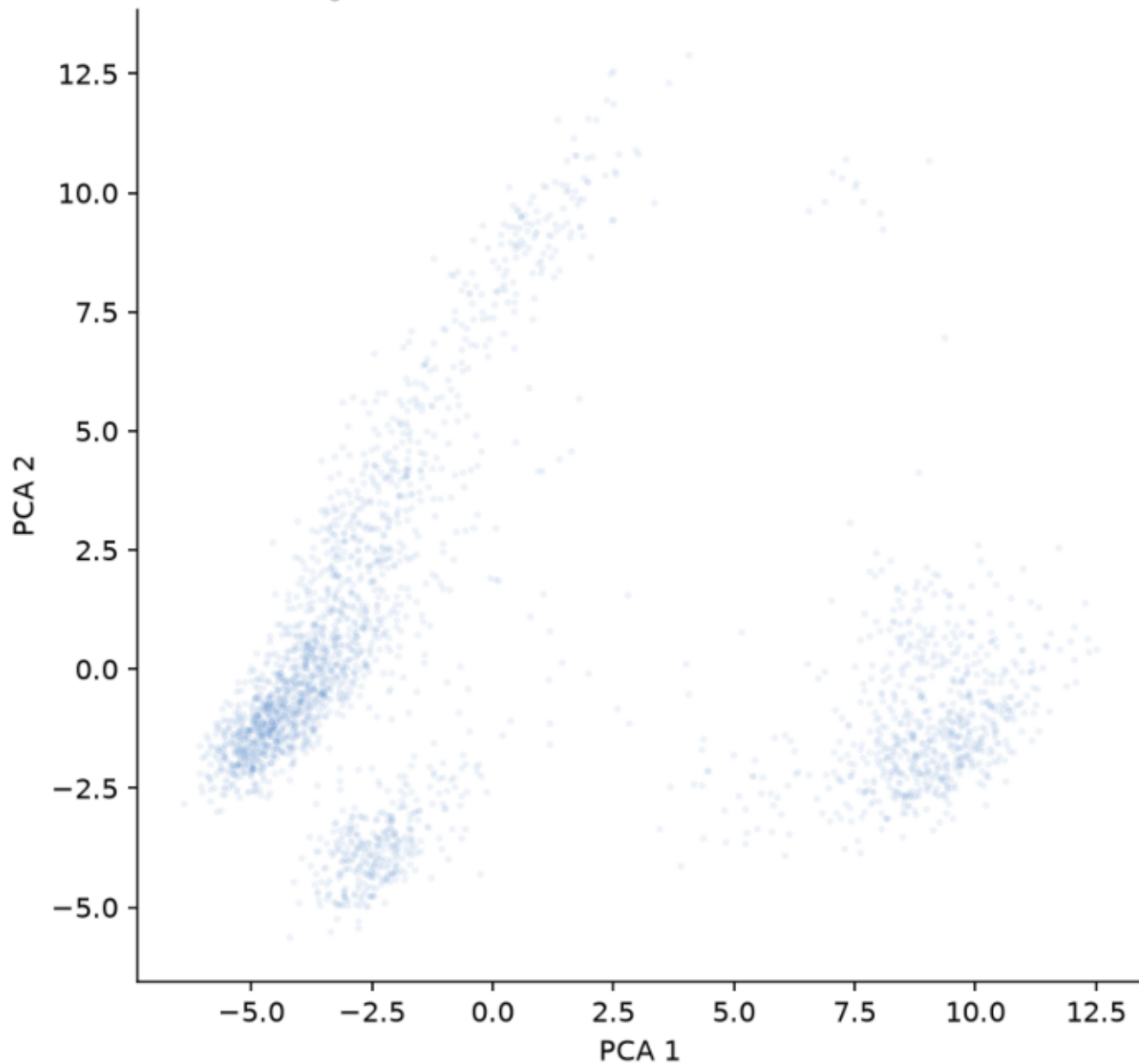
The fraction of the total variance in the preprocessed data captured by the principal components actually kept in the embedding: the sum of those components' explained-variance ratios (each component's variance divided by the total). Ranges 0 to 1. PCA only; blank for methods it does not apply to.

PCs for 90% variance

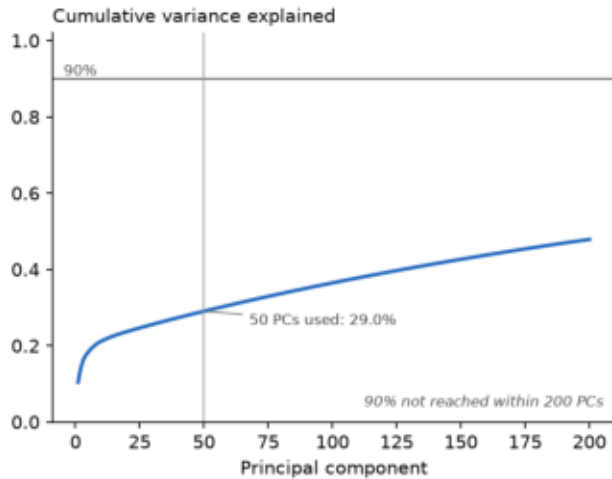
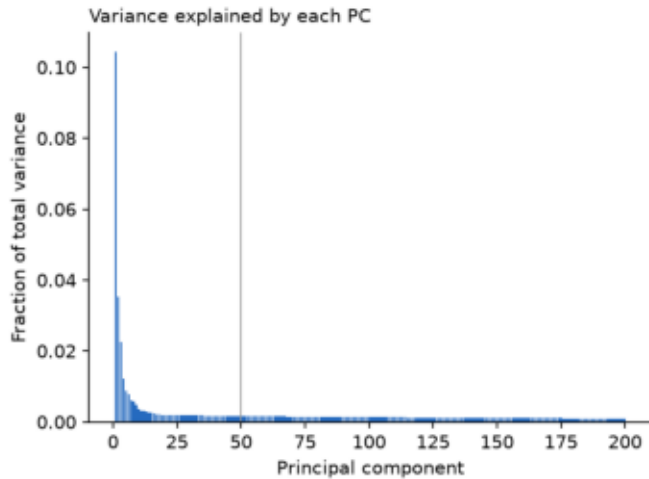
The smallest number of principal components whose cumulative explained variance reaches 90%, read off a diagnostic PCA fit of up to $\min(4 \times n_{\text{components}}, 200)$ components. A leading '>' means 90% was not reached within that fit, so the true number is larger than shown. The scree plots show the full curve.

pbmc: pca

n=2,700, showing dims 1-2 of 50

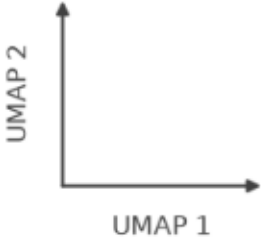


pbmc: pca scree plot



pbmc: umap

n=2,700



Findings and interpretation

Method results

PCA (general purpose, 50 components) has trustworthiness 0.9344 and captures 0.2904 of the total variance. 90% of the variance was not reached within the 200-component diagnostic fit (more than 200 PCs are needed). The variance spectrum is flat, so variance in this dataset is spread across many directions rather than concentrated in a few, and a linear projection to 50 dimensions keeps under a third of it.

UMAP (visualization only, 2 components) has trustworthiness 0.7740. This is not a like-for-like comparison with PCA's 0.9344, because the two embeddings have 2 and 50 dimensions; a 2-dimensional layout is expected to preserve less neighbourhood structure than a 50-dimensional one. The two methods do different jobs: PCA is the numeric representation, UMAP is the picture.

What the figures show

The UMAP plot shows three visually separated groups and one isolated point. The PCA plot, in its first two components, shows a more continuous and less separated cloud. This dataset has no labels, so the groups cannot be checked against known cell types, and UMAP's group sizes and the gaps between groups should not be read as measurements. Treat the structure as exploratory.

Limitations

No ground-truth labels available for this dataset; embedding quality assessed only via label-free trustworthiness, no supervised sanity check possible.

No adversarial critique/revision pass was performed for this run (critic subagent not yet built).